

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

C02: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology.
(BL3)

Tool Weightage: 30%

Product:

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(Name / Reg No) certify that this submission is my original
that I have adhered to the guidelines specified for this assessment.
that any violation of academic integrity rules will result in disciplinary

S. J.

the student:

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Understanding of Scenario	20%	1.4	1.4	1.6	1.6	1.
Identification of Key Issues	20%	1.6	1.6	1.4	1.4	1
Application of Concepts	25%	1.	1	1	1	1.6
Decision Making	20%	1	1	1	1	1
Clarity of Response	15%	0.4	1	1	1	0.4

28/50 22/5

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → /24

Network

block: $192.168.100.0/24$

1) School of computing - /25

Network address: $192.168.100.0$ Broadcast address: $192.168.100.127$ Valid: $192.168.100.1 - 192.168.100.126$

Usable hosts: 126.

2) School of Electronics - /26

The next available address after 127 is $192.168.100.128$ ⇒ Network: $192.168.100.128$ ⇒ Broadcast: $192.168.100.191$ ⇒ Valid: $192.168.100.129 - 192.168.100.190$

Usable host: 62.

3) School of Mechanical Engineering: - /27

The next available address is $192.168.100.192$.Next address: $192.168.100.192$.Broadcast: $192.168.100.223$ Valid: $192.168.100.193 - 192.168.100.222$.

Usable hosts: 30.

4) Remaining unused IP range.

The original /24 block ends at $192.168.100.255$ Allocated up to $192.168.100.223$

Therefore, remaining unused range::

$$192.168.100.224 - 192.168.100.255$$

This contains 32 addresses, including one network and one broadcast address. It is treated as a /27 block.

2. Multinational software company - Subnet Allocation.
Given

IP block: 10.10.0.0/16

Required subnets.

1) Software Development - /24

A /24 subnet contains ~~256~~ addresses

⇒ Network address: 10.10.0.0

⇒ Broadcast address: 10.10.0.255

⇒ valid host range: 10.10.0.1 - 10.10.0.254

⇒ Usable hosts: 254

2) Quality Assurance - /25

⇒ Next available address: 10.10.1.10

⇒ Broadcast address: 10.10.1.127

⇒ valid host range: 10.10.1.1 - 10.10.1.126

⇒ Usable hosts: 126.

3) Human Resource: - /26.

Next available address: 10.10.1.128

⇒ Network address: 10.10.1.128

⇒ Broadcast address: 10.10.1.191

usable host range: 10.10.1.129 - 10.10.1.190
usable hosts: 62.

4) Executive Management - /27

Next available address: 10.10.1.192

⇒ Network address: 10.10.1.223

⇒ valid host range: 10.10.1.192 - 10.10.1.222

⇒ usable host: 30.

5) unused IP address Range.

The allocated block is 10.10.0.0/16, which extends from.

10.10.0.0 - 10.10.255.255.

The assigned subnets end at 10.10.1.223.

Therefore the remaining unused range is

10.10.1.224 - 10.10.255.255

3) Smart Hospital - IPv6 Addressing

Given .

2001:0db8:abcd:0000:0000:0000:0000:0000/64

Compressed IPv6 Address

2001:db8:abcd::/64

Expanded IPv6 address:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

Network prefix (/64)

⇒ Network prefix: 2001:db8:abcd::/64

⇒ The first 64 bits represent the network and the last 64 bits are used for host addresses

Total no of Host addresses:

For a /64 prefix

$$2^{128-64} = 2^{64}$$

Total host addresses = 10,446,744,073,709,551,616

Total possible Host Addresses Available :

⇒ 18,446,744,073,709,551,616 (2^{64}) address.

4) ISP Routing Table :

Given

→ 192.168.1.0/24

→ 192.168.2.0/24

→ 192.168.3.0/24

Destination IP Address : 192.168.2.45

1) determine the Destination Subnet

The destination IP address is 192.168.2.45 since routing table contains the subnet 192.168.2.0/24 this belongs to that network.

→ Network Address : 192.168.2.0

→ Subnet Mask : 255.255.255.0/24

→ valid Host Range : 192.168.2.1 - 192.168.2.254

→ Broadcast Address : 192.168.2.255

As 192.168.2.45 lies within this range, the destination subnet is 192.168.2.0/24

2) Identify the Matching Routing table Entry

The router compares the destination IP with

Routing table entries using the longest match rule.

Routing Entry:

	Match
192.168.1.0/24	No
192.168.2.0/24	Yes
192.168.3.0/24	No

Therefore the matching routing table entry is

192.168.2.0/24

a) Determine the Next-Hop Network/Interface.

After finding the matching route, the router forwards the packet through the interface.

5) Company department LAN subnetting:

Given:

Administration LAN: 192.168.10.0/26

Finance LAN: 192.168.10.64/26

Router Gateways:

→ Administration Gateway: 192.168.10.1

→ Finance Gateway: 192.168.10.65

Administration LAN (192.168.10.0/26)

Subnet Mask: 255.255.255.192

Network Address: 192.168.10.0

Broadcast Address: 192.168.10.63

Finance LAN:

Subnet mask: 255.255.255.192

Network Address : 192.168.10.64

Broadcast Address : 192.168.10.127

a) Identify the valid Host IP Address Range

Administration LAN

→ First Host : 192.168.10.1

→ Last Host : 192.168.10.62

→ Valid Host Range : 192.168.10.1 - 192.168.10.62

→ Usable Hosts - 62.

Finance LAN:

→ First Host : 192.168.10.65

→ Last Host : 192.168.10.126

→ Valid Host Range : 192.168.10.65 - 192.168.10.126

→ Usable Host : 62.